Predicting SNAP Payment Errors: A Pilot Proposal for South Dakota

Synthetic-data-validated case prioritization for quality-control review — tested where errors are rarest

October 4, 2026 · Research whitepaper · Methods and validation artifacts are open-source at github.com/ctrimm/snap-qc-per-experiments

1. Executive summary

South Dakota has the lowest SNAP payment error rate in the nation (3.28% FY2024, federally published). That makes it the hardest and most valuable test of our approach: if synthetic-data-trained models can find errors *where errors are rare*, the method is robust. We built and validated a South Dakota-conditioned synthetic data generator, calibrated against five years of federal quality-control (QC) microdata (2,365 SD cases, FY2020–2024). Machine-learning models trained *only* on this synthetic data rank real South Dakota cases by error risk well above chance — finding AUC (a ranking metric where 0.5 is chance and 1.0 is perfect) of 0.750 vs 0.680 for a nationally trained model — and the advantage survived a fully clean refit that excluded every holdout row. Proof the synthetic cases carry genuine signal about where errors happen, even in the nation's cleanest caseload.

The operational finding — and the track's biggest surprise: ranking cases by error probability *alone* is the winning rule, recovering **$2,644 vs $1,275 for random review** at a fixed review budget — **$45 vs $22 per hour of reviewer time**. This reverses the Pennsylvania and New York pattern, where probability-alone was worse than random and dollar-aware rankers won. Reviewing the riskiest half of cases catches 76% of all errors with half the reviews.

We propose a low-risk pilot: train the ranker on South Dakota's own historical QC data, validate on a recent holdout, then run it in shadow alongside existing review. No production eligibility-system changes. No new data collection.

2. The problem

SNAP payment error rates remain stubbornly high nationwide, and every error is either taxpayer money over-issued or a household underpaid. Quality-control review is the corrective mechanism — but reviewer time is finite, and cases are currently selected for review without a data-driven sense of where errors concentrate. The result is expensive review effort spread thinly over a caseload where risk is highly uneven.

South Dakota is the extreme test of a targeted approach. Our analysis of 2,365 South Dakota QC cases (FY2020–2024) shows the state's benefit-calculation engine is the cleanest of the three states studied (99.0–100% of cases match federal statute exactly, with zero computation variances in five years) — South Dakota's errors live entirely in the *human process layer* (verification, timing, documentation), which is precisely where a prediction tool adds value. The error signature is distinctive: timing errors — cases the agency was not yet required to act on — run at 1.89x the national rate and make up three-quarters of all errors, the most timing-heavy of the three states studied; earned-income errors run 1.45x and medical-deduction errors 1.40x, while shelter and utility-allowance errors run below national rates. Error dollars are small (median $17 vs $28 nationally) and hyper-concentrated: the top tenth of errors by dollar size holds 83% of all error dollars.

3. What was built and validated

A South Dakota-conditioned synthetic generator. Starting from a nationally calibrated generator, we fit South Dakota-specific error-rate offsets and empirical error mixes. The SD-conditioned output reproduces South Dakota's finding mix (L1 0.004 — the tightest of the three states), element mix, and cause mix, while leaving national output byte-identical. A bespoke calibration fix (rebasing cause ratios against the generator's mixed cause basis) was needed because South Dakota records almost no agency-fault causes on income errors — a state-specific quirk the national recipe mishandled.

Validated predictive signal — the strongest of the three states. Models trained only on SD-conditioned synthetic cases predict real, held-out South Dakota errors better than the nationally trained model on every tested seed: finding AUC 0.728 vs 0.638 (seed 7), and on element classification the SD model wins macro-F1 (a per-class balanced accuracy metric) on all seeds. The honest threat — whether the generator had indirectly seen the holdout data — was tested with a fully clean refit on 1,773 rows excluding every holdout case: the advantage not only survived but measured slightly higher (AUC 0.750 vs 0.680). There is no contamination premium.

Probability-alone wins — the PA/NY reversal. In Pennsylvania and New York, ranking by error probability alone was *worse than random* at the top slice, because probability and error size were

nearly orthogonal: the dollars lived in a big-dollar income tail that probability couldn't see. South Dakota has no big-dollar tail (90th percentile: $70), and the synthetic dollar regressors are miscalibrated (they predict more dollars than they recover) — so multiplying a good probability ranker by a bad dollar estimate *hurts*. Probability-alone is the best rule at the top-10% and top-30% review budgets and ties at top-20%. The lesson for deployment: the ranking rule must be re-fit to each state's error economics, not copied between states.

Headline pilot economics (simulated on held-out SD cases, top-20% review budget):

	Random review	P(error)-only ranker
Error dollars recovered	$1,275	**$2,644**
Value per review hour	$22	**$45**
Share of all error dollars captured	—	41.3%

Of the $2,644: $1,884 is over-issuance (direct program savings); $760 is under-issuance (households underpaid — the correct QC outcome, but a make-whole cost). Savings-only value: $31.93/hr vs $13.48/hr for random review. Absolute dollars are small — South Dakota's errors are rare *and* small — but the lift over random (2.07x) is the largest of the three states studied.

4. The pilot

What South Dakota provides

- Historical QC case extracts the state already holds (no new data collection, no new PII).
- A review team willing to work a ranked queue for a defined pilot period.

What changes — and what does not

- **Phase 0 (cold start, no SD data needed):** the synthetic-trained ranker scores a sample of cases; rankings are validated against known outcomes.

- **Phase 1:** the ranker is trained on SD's historical QC data and validated on a recent holdout the model never saw. Phase 1 also selects the ranking rule for SD's economics (probability-alone vs dollar-aware) rather than inheriting another state's.
- **Phase 2 (shadow):** the ranker scores the live review caseload; reviewers work the top slice; outcomes are measured against the baseline selection method.
- **What does not change:** no modifications to the eligibility system, no automated determinations (the model prioritizes review — it never decides a case), no data leaves the state's control.

Success metrics

- Precision in the reviewed slice vs baseline selection.
- Error dollars recovered per reviewer hour vs baseline.
- Reviewer time saved to reach a fixed error-capture target.

A reasonable pilot shape is 8–12 weeks from data handoff to shadow-readout.

5. Limitations and honest caveats

- **Small samples, wide intervals.** The holdout contains only 152 error cases; per-seed margins are thin and confidence intervals are wide. We report three seeds and prefer qualified claims — the finding-task transfer is consistent across seeds, but this is a small-sample result and should be re-validated on SD's own larger historical extract in Phase 1.
- **The ranking rule is state-specific.** Probability-alone wins in South Dakota because there is no big-dollar tail and the dollar regressors are miscalibrated. If SD's error economics shift (a big-dollar tail emerging), the rule must be re-tuned — Phase 1 selects the rule per state rather than assuming transfer.
- **Refit annually.** The transfer signal is year-to-year stable, but the underlying mix is shifting: the case-error rate rose from 21.6% to 29.4% (FY2020–22 vs FY2023–24), and FY2024 brought a coding-practice change (the state began recording causes on timing errors, shifting the cause mix sharply). Shelter-deduction errors showed measurable decay and should be watched. Partial-year files (FY2020–21) are excluded from refits.
- **Procedural errors are invisible.** The federally published payment error rate (3.28% FY2024) and the case-microdata-implied case-error rate (24.75%) measure different things:

the published rate is dollar-weighted and includes procedural errors the microdata cannot see. This tool targets substantive case errors, not procedural compliance.

- **Headroom.** The synthetic-trained ranker captures roughly three-quarters of the performance a model trained on real SD data is projected to reach (2.07x vs 2.82x lift). The pilot's Phase 1 exists to close exactly this gap.
- **Human judgment stays in the loop.** This is a triage tool for reviewers, not an adjudication engine. Final determinations remain human — by design.

Review-burden note

In the Pennsylvania and New York tracks, dollar-maximizing rankers proved risk-regressive: they deprioritized elderly and disabled households (the highest-error groups) because their errors are frequent but small-dollar. South Dakota's winning rule is probability-alone, which does *not* chase dollars — so this particular regressivity mechanism does not apply to the recommended SD configuration. If a future Phase 1 selects a dollar-aware rule for SD, the type-stratified review-burden analysis from the PA/NY tracks should be re-run on SD data before deployment.

6. The ask

We are looking for a South Dakota partner to run this pilot: a data contact who can provide historical QC extracts, and a review team for the shadow phase. We bring the validated methodology, the synthetic cold-start model, and the open-source validation artifacts. All methods, experiments, and results are published openly; nothing about the approach is proprietary or black-box.

Methods, experiment chronicles, and validation code: github.com/ctrimm/snap-qc-per-experiments (private research repo; access on request). This work is conducted in the open-source civic-technology lane. Contact: via the repository.